

Causal Structure in Galaxies: Comparing Simulations and Observations

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ABSTRACT

Cosmological simulations are calibrated to reproduce galaxy scaling relations, yet two models can match the same scaling relations while differing in their causal structure. I use causal discovery, via the Fast Causal Inference with Targeted Testing (FCIT) algorithm, to estimate this structure as partial ancestral graphs for two observational catalogues (the NASA–Sloan Atlas and the gas-selected ALFALFA–NSA sample) and three independent cosmological simulations: TNG50, EAGLE, and SIMBA. Only the simulations provide true halo and intrinsic masses, and three independent codes separate universal from subgrid physics. Of the 56 relations recovered across the three codes, only eight are invariant—present in all three with the same direction—led by the stellar mass–velocity dispersion (Faber–Jackson-like) relation; this backbone is 4–8 edges depending on the penalty and is largely fixed by the variable definitions. A within-code null test separates two effects: over 40 random partitions the codes recover substantially different *sets* of relations (skeleton overlap median 0.33 across codes versus 0.79 within; permutation $p < 0.001$), yet the directions FCIT assigns are *accurate*—85 per cent correct against latent-free mock graphs with known structure (chance 50 per cent). The model-dependence therefore lies mainly in which relations a simulation recovers, not their direction, making causal structure a diagnostic that flags where galaxy-formation models diverge.

Keywords: Galaxy evolution (594) — Galaxy formation (595) — Hydrodynamical simulations (767) — Astrostatistics (1882) — Astronomy data analysis (1858) — Galaxy dark matter halos (1880)

1. INTRODUCTION

The description of the properties of galaxies is built around the set of correlations between galaxy parameters: bigger masses correspond to bigger sizes (Shen et al. 2003; van der Wel et al. 2014) and higher metallicities (Tremonti et al. 2004), mass determines the kinematical properties of galaxies (Tully & Fisher 1977), and the amount of gas, colour, and star formation are tightly related (Brinchmann et al. 2004; Gallazzi et al. 2005). All these relations serve as a standard for validating the galaxy-formation models. However, a correlation says nothing about what parameter controls another, and whether they are directly connected or there is some other latent variable that controls both. This question is crucial in discussions about quenching mechanisms that may depend on stellar mass, halo mass, or central black hole mass (Bluck et al. 2020). The unobserved dark-matter halo mass (Behroozi et al. 2019) and dust attenuation (Calzetti et al. 2000), as well as selection

effects such as a flux limit that correlates distance and luminosity (Teerikorpi 2015), adds complexity to this task.

Causal discovery tries to answer the questions above explicitly. Based on the graphical-modelling framework of Pearl (2009) and Spirtes et al. (2000), it detects direct dependencies from data, estimates the direction of the dependency if possible, and detects the existence of latent confounders, encoding all this information as a partial ancestral graph (PAG). Recently, causal and causal-discovery methods have been applied to astrophysical problems (Pasquato et al. 2023; Jin et al. 2025; Mucesh et al. 2024; Bluck et al. 2022); in particular, Desmond & Ramsey (2026) applied Fast Causal Inference with Targeted Testing (FCIT) (Ramsey et al. 2025a,b) to the NASA–Sloan Atlas and obtained a mass-based hierarchical structure. They point out that only in comparison with cosmological simulations, where the input physics is known, can one distinguish the physical structure from the selection and measurement biases.

That is why the comparison between the results of causal discovery based on real data and those based on

a simulation becomes the goal of this study. The problem is that in the real data some decisive variables are missing (such as halo properties or the intrinsic star-formation rate) or obtained from other variables (for instance, stellar mass obtained from photometry lies downstream of luminosity). In a simulation, by contrast, all parameters are available at the ground-truth level, serving as a natural laboratory for causal discovery. Nevertheless, one simulation is not enough, because the structure it provides may be a product of the subgrid physics of that particular simulation rather than galaxy physics itself. Thus, I use three simulations. I apply FCIT to two observational catalogues—the NSA (Blanton et al. 2011) and the gas-selected ALFALFA–NSA sample (Stiskalek et al. 2021; Haynes et al. 2018)—and three state-of-the-art cosmological simulations: TNG50 (Pillepich et al. 2018; Nelson et al. 2019a), EAGLE (Schaye et al. 2015; McAlpine et al. 2016), and SIMBA (Davé et al. 2019). This identifies the relationships that make up the invariant backbone reconstructed by all three codes (candidate robust physics), determines those whose orientation is dictated by a particular subgrid model, and introduces causal structure as a discriminator between models that the common scaling laws cannot provide.

2. DATA

The analysis is based on five datasets: two observational catalogues—the ALFALFA–NSA matched sample (gas-selected) and the broader NSA catalogue (optical)—and three cosmological simulations built on independent galaxy-formation models, TNG50, EAGLE, and SIMBA. The per-sample causal graphs of Section 4 are presented for the observational samples and TNG50, while the multi-code comparison of Section 5 uses all three simulations, and the observations, on matched variables.

For each dataset I selected a core set of first-order, widely measured structural, photometric, and mass-related galaxy properties, and applied consistent quality cuts. For each sample I included those properties that are reliably available in it: the gas observables (HI mass and line width) therefore appear only for the ALFALFA–NSA sample, and the halo, black-hole, and kinematic quantities only for TNG50. Property names prefixed `ELPETRO` denote elliptical Petrosian measurements from the NSA. Unless stated otherwise, masses are quoted as $\log_{10}(M/M_{\odot})$. The quality cuts listed below bracket the range over which each quantity is reliably measured (or, for TNG50, well resolved), removing catalogue artefacts and poorly resolved or unconverged objects. Hyperparameter optimisation was performed

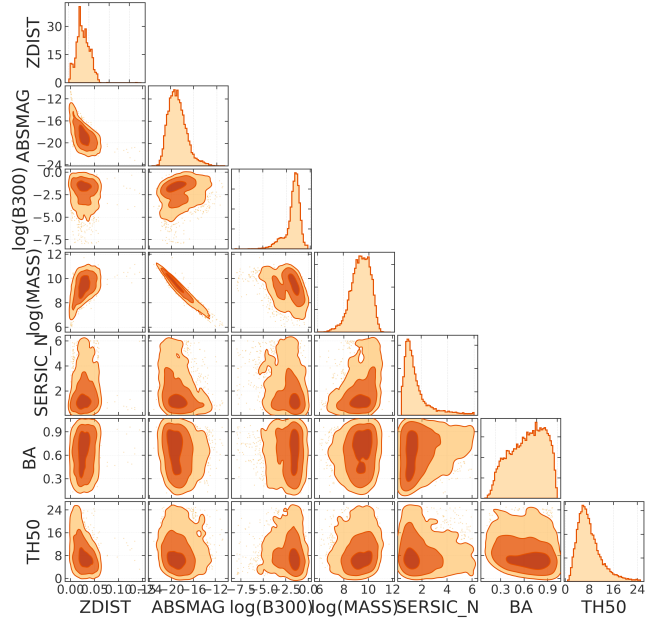


Figure 1. Corner plot of the final cleaned ALFALFA–NSA matched sample. Contours contain 39.3, 86.5 and 98.9 per cent of the points (1, 2 and 3σ).

separately on each sample to choose the best truncation limit and penalty discount for FCIT. Corner plots of the three cleaned samples are shown in Figs 1–3; for clarity these show a representative subset of the properties, while the full variable lists and quality cuts are given in Tables 1–3. Because the three samples are selected very differently—a gas-rich, blue, flux-limited population (ALFALFA–NSA), the broader optical population (NSA), and a simulated population (TNG50)—their marginal distributions of quantities such as colour and metallicity differ, driven by the selection functions and, for TNG50, by the simulation’s subgrid physics rather than by the causal-discovery procedure.

2.1. ALFALFA–NSA Matched Sample

The ALFALFA–NSA matched sample combines 21 cm HI (neutral-hydrogen) detections from the ALFALFA (Arecibo Legacy Fast ALFA) survey with optical and near-infrared photometry from the NASA–Sloan Atlas (NSA) (Stiskalek et al. 2021; Haynes et al. 2018).

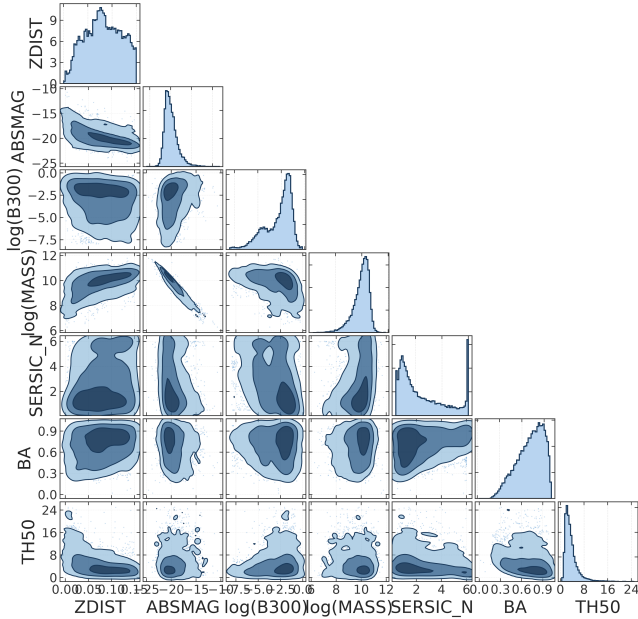
The original catalogue contains 31,502 galaxies; after quality cuts the final sample contains 20,569 gas-rich, blue galaxies at low redshift. This catalogue is flux-limited and gas-selected, preferentially including galaxies with high HI content that are detectable in radio surveys.

The properties used are listed in Table 1.

Quality cuts are redshift < 0.15 , stellar mass 10^6 – $10^{12} M_{\odot}$, HI mass 10^6 – $10^{10.5} M_{\odot}$, HI line width 20–

Table 1. Properties used in the ALFALFA–NSA matched sample analysis.

Property	Description
ZDIST	Estimated cosmological redshift corrected for peculiar velocity (Mpc/h)
ELPETRO_ABSMAG_R	Absolute magnitude in the SDSS <i>r</i> -band
ELPETRO_B300	Current star formation rate divided by the average over the past 300 Myr
ELPETRO_MASS	Stellar mass ($\log_{10} M_{\odot}$)
SERSIC_N	Sérsic index from a 2D single-component fit in the <i>r</i> -band (morphology)
ELPETRO_BA	Axis ratio b/a at the isophotal contour enclosing 90% of the light
ELPETRO_TH50_R	Angular half-light radius in the <i>r</i> -band (arcseconds)
logMH	HI mass ($\log_{10} M_{\odot}$)
W50	HI line width at 50% peak (km s^{-1})
BARYONIC_MASS	Baryonic mass (stellar mass + $1.4 \times$ HI mass; derived)
COLOR_U_R	$U - R$ colour (derived from photometry)
ELPETRO_METS	Stellar metallicity
ELPETRO_MTOL	Mass-to-light ratio in the <i>r</i> -band

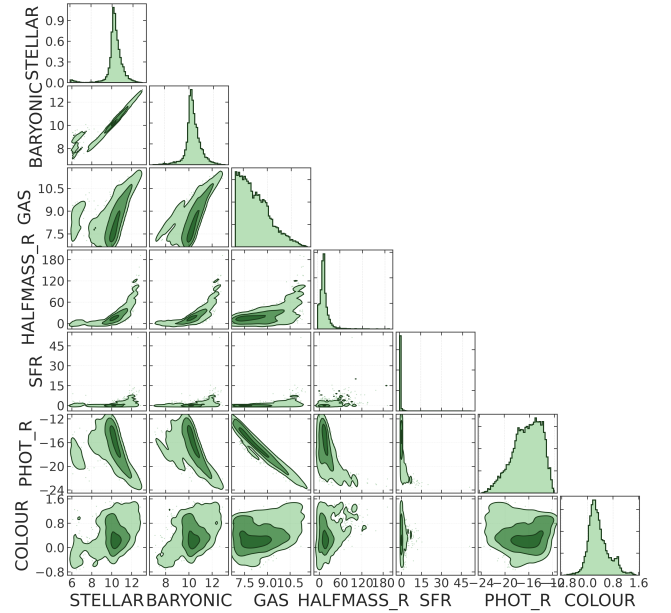
**Figure 2.** Corner plot of the final cleaned NSA sample. Contours contain 39.3, 86.5 and 98.9 per cent of the points (1, 2 and 3σ).

500 km s^{-1} , and half-light radius 0–25 arcsec. These cuts remove objects with unreliable photometry or HI fluxes while preserving the gas-rich nature of the sample.

2.2. NSA Sample

The NSA catalogue provides a broader view of the low-redshift galaxy population. It is drawn from Sloan Digital Sky Survey (SDSS) imaging and spectroscopy and contains 641,409 galaxies with robust stellar masses, sizes, colours, star-formation rates, and morphological parameters (Blanton et al. 2011).

For this analysis I applied quality cuts for reliable measurements across all required properties; the SDSS

**Figure 3.** Corner plot of the final cleaned TNG50 sample. Contours contain 39.3, 86.5 and 98.9 per cent of the points (1, 2 and 3σ).

spectroscopic parent is flux-limited. The final sample contains 484,539 galaxies.

The properties used are listed in Table 2.

Quality cuts are redshift < 0.15 , stellar mass $10^6 - 10^{12} M_{\odot}$, absolute *r*-band magnitude between -25 and -10 , Sérsic index < 6 , axis ratio between 0 and 1, and half-light radius 0–25 arcsec. These cuts follow standard NSA cleaning procedures and remove objects with unreliable photometry or spectroscopy while keeping a representative sample of the local population.

Table 2. Properties used in the NSA sample analysis.

Property	Description
ZDIST	Estimated cosmological redshift corrected for peculiar velocity
ELPETRO_ABSMAG_R	Absolute magnitude in the SDSS r -band
ELPETRO_B300	Current star formation rate divided by the average over the past 300 Myr
ELPETRO_MASS	Stellar mass ($\log_{10} M_{\odot}$)
SERSIC_N	Sérsic index from a 2D single-component fit in the r -band
ELPETRO_BA	Axis ratio b/a at the isophotal contour enclosing 90% of the light
ELPETRO_TH50_R	Angular half-light radius in the r -band (arcseconds)
COLOR_U_R	$U - R$ colour (derived from photometry)
ELPETRO_METS	Stellar metallicity
ELPETRO_MTOL	Mass-to-light ratio in the r -band

2.3. TNG50 Sample

The TNG50 run of the IllustrisTNG project (Pillepich et al. 2018; Springel et al. 2018; Nelson et al. 2018; Marinacci et al. 2018; Naiman et al. 2018; Nelson et al. 2019b; Pillepich et al. 2019) provides a high-resolution cosmological hydrodynamical simulation in which every important galaxy property is known; the simulation has been extensively validated against observed galaxy populations, including their optical morphologies (Rodríguez-Gomez et al. 2019), and the data are publicly available through the IllustrisTNG release (Nelson et al. 2019a).

I selected central subhaloes at redshift zero with well-resolved stellar and gas components; after quality cuts the final sample contains 10,992 galaxies that span a wide dynamic range in mass, size, metallicity, and kinematics.

The properties extracted are listed in Table 3.

Quality cuts are dark-matter mass $> 10^5 M_{\odot}$, stellar mass 10^6 – $10^{14} M_{\odot}$, gas mass $10^{6.5}$ – $10^{11.5} M_{\odot}$, black-hole mass $< 10^{12} M_{\odot}$, half-mass radius between 0.03 and 316 kpc, velocity dispersion 2–300 km s^{-1} , maximum circular velocity 5–500 km s^{-1} , and star-formation rate $< 316 M_{\odot} \text{ yr}^{-1}$. These cuts remove poorly resolved or unphysical objects while preserving the full dynamic range of the simulation. I note, however, that roughly two-thirds of the selected subhaloes contain no resolved central black hole; for these objects BH_MASS is assigned the catalogue floor value ($\log_{10}(M_{\text{BH}}/M_{\odot}) = -10$), producing a strong pile-up at that value. Edges involving black-hole mass in the TNG50 graph should therefore be treated as tentative, as they are partly shaped by this bimodal distribution rather than by a single continuous physical relation.

2.4. EAGLE Sample

EAGLE (Schaye et al. 2015; Crain et al. 2015) is a cosmological hydrodynamical simulation run with a modified version of GADGET-3 (smoothed-particle hy-

drodynamics), with thermal stellar and active galactic nucleus (AGN) feedback calibrated to the $z = 0$ stellar-mass function, galaxy sizes, and the black-hole–stellar-mass relation. I use the reference run RefL0100N1504 (a 100 comoving-Mpc box) at $z = 0$, querying the public database (McAlpine et al. 2016) for central galaxies. For each galaxy I extract 30-kpc-aperture stellar and gas mass, star-formation rate and velocity dispersion, together with the dark-matter (subhalo) and black-hole mass, stellar half-mass radius, stellar and star-forming-gas metallicity, and dust-free SDSS u - and r -band magnitudes. After the same finiteness and resolution cuts applied to the other simulations, the sample contains 11,887 central galaxies.

2.5. SIMBA Sample

SIMBA (Davé et al. 2019) is built on the GIZMO meshless-finite-mass solver, with a galaxy-formation model independent of both IllustrisTNG and EAGLE: torque-limited black-hole accretion, two-mode (kinetic and jet) AGN feedback, and an on-the-fly dust model. I use the public flagship run (m100n1024; a 100 $\text{Mpc } h^{-1}$ box) at $z = 0$, reading galaxy properties from its CAESAR catalogue (Thompson 2015). The same first-order properties used for TNG50 and EAGLE are extracted, with dust-free synthetic SDSS photometry; the dark-matter halo mass is taken from each galaxy’s parent halo. Selecting central galaxies yields 32,757 objects after the standard cuts; as in TNG50, galaxies with no resolved central black hole are assigned the catalogue floor.

2.6. Matched samples for the cross-catalogue comparison

The per-sample graphs are recovered on each catalogue’s own variable set and so cannot be compared directly. To compare causal structure *across* the simulations I build a matched sample restricting TNG50, EAGLE, and SIMBA to the full intrinsic set all three

Table 3. Properties used in the TNG50 sample analysis.

Property	Description
DM_MASS	Dark-matter halo mass ($\log_{10} M_{\odot}$)
STELLAR_MASS	Stellar mass ($\log_{10} M_{\odot}$)
GAS_MASS	Gas mass ($\log_{10} M_{\odot}$)
BH_MASS	Black-hole mass ($\log_{10} M_{\odot}$)
BARYONIC_MASS	Baryonic mass (stellar + gas)
HALFMASS_RAD	Half-mass radius (kpc)
VEL_DISP	Velocity dispersion (km s^{-1})
VMAX	Maximum circular velocity (km s^{-1})
GAS_METALLICITY	Gas-phase metallicity
STAR_METALLICITY	Stellar metallicity
PHOTOMETRIC_R	r -band absolute magnitude
PHOTOMETRIC_U	u -band absolute magnitude
COLOUR	$U - R$ colour
SFR	Star-formation rate ($\log_{10} M_{\odot} \text{ yr}^{-1}$)

provide: dark-matter, stellar, gas, black-hole and baryonic mass, size, velocity dispersion, stellar and gas-phase metallicity, r - and u -band magnitude, colour, and star-formation rate. Every variable is standardised within each sample, heavy-tailed positive quantities (size, SFR) are log-scaled first, and each sample is randomly subsampled to a common $N = 10,000$ galaxies, so that differences between the recovered graphs cannot be attributed to sample size, nor to the hyperparameter and variable-set differences that distinguish the per-sample runs. All three simulations are processed with the same pipeline, so the resulting matched datasets differ only in their subgrid physics.

3. METHODS

Partial Ancestral Graphs (PAGs) were reconstructed using the Fast Causal Inference with Targeted Testing (FCIT) algorithm on the NSA and ALFALFA-NSA observational catalogues and the TNG50, EAGLE, and SIMBA cosmological simulations.

The aim is to recover the causal structure in each catalogue and then to compare it across the observations and the three independent simulations, as set out in Section 5.

3.1. FCIT

FCIT is a hybrid constraint-based and score-based approach for large astrophysical data sets that may display nonlinearity and latent confounding (Ramsey et al. 2025a).

It follows a three-step process. First, an undirected skeleton is found by checking if two variables are conditionally independent given a small subset of conditioning variables. The benefit of localised conditioning is that the number of tests decreases compared to previ-

ous approaches where all possible variable conditioning is checked.

Second, targeted blocking is used to solve any remaining dependencies which cannot be addressed using ordinary conditioning. Third, orientation rules are applied to obtain a PAG where directed edges indicate a clear cause-effect relation and circle marks show potential ambiguities caused by latent variables or orientation of causation (Ramsey et al. 2025a; Zhang 2008).

Independence testing in FCIT uses the basis-function likelihood ratio test. To check if $X \perp Y \mid Z$, the candidate variable X and the conditioning set Z are expanded in terms of Legendre basis functions up to the truncation limit. Conditional independence is then tested against a restricted model

$$H_0 : Y = f(Z) + \epsilon \quad (1)$$

and unrestricted model

$$H_1 : Y = f(Z) + g(X) + \epsilon, \quad (2)$$

where f and g are the respective basis-function expansion models and ϵ is Gaussian noise (Ramsey et al. 2025b).

Basis-function Bayesian Information Criterion (BIC) is used for model selection:

$$\text{BIC} = L - \text{penalty_discount} \times k \ln(N), \quad (3)$$

where L is the maximised log-likelihood, k is the number of coefficients, and N is the local sample size. This enables FCIT to detect the strong nonlinearity that occurs in galaxy data while staying computationally feasible even when working with several thousands of objects. In practice, FCIT was run through the PY-TETRAD inter-

face (Ramsey & Andrews 2023) to the TETRAD causal-discovery package, using the basis-function likelihood-ratio test at significance level $\alpha = 0.01$ together with the basis-function BIC described above.

3.2. Hyperparameter Optimisation

In order to obtain robust results, hyperparameter tuning was done independently for each data set. Truncation limit determines the complexity of basis function expansions in the independence test calculations, whereas the penalty discount controls the influence of model complexity on the BIC score.

For the TNG50 and ALFALFA–NSA data sets, 100 synthetic mock data sets with a pre-defined causal structure were generated using the Causal Perceptron Network (CPN) framework (Ramsey et al. 2025b).

Each mock data set was generated through the process of creating a directed acyclic graph (DAG) with 12–16 edges, defining non-linear dependencies between the variables through a multi-layer perceptron, and adding independent noise distributed according to Beta(2,5) distribution. After an initial exploration of different truncation limits, a fixed value of truncation limit of 7 was chosen, and then the penalty discount was tuned by grid search over the values {5, 8, 12, 15, 20, 25, 30, 35, 40, 50, 75, 100, 150, 200}, selecting the value that maximised the mean F1 score—the harmonic mean of edge precision and recall, evaluated against the known mock structure—across the 100 mock data sets. These mocks reproduce the dimensionality, sample size, and approximate sparsity of each real data set, but are not calibrated to its specific correlation structure or measurement noise; the calibration therefore targets generic nonlinear, latent-free data of the appropriate scale, and the hyperparameter-sensitivity analysis on the real data described below provides the complementary check that the recovered structures do not depend strongly on the chosen values.

In the resulting grid search for TNG50, the mean F1 score was 0.811 at penalty discount = 15, and in the case of ALFALFA–NSA, the mean F1 score was 0.805 at penalty discount = 35. In terms of the NSA catalogue, however, there was no further re-fitting of the hyperparameters. The values of the truncation limit (= 14) and penalty discount (= 50) were taken straight from Desmond & Ramsey (2026), who used FCIT on the same NSA catalogue. Thus, the NSA graph is obtained with externally calibrated hyperparameters rather than ones tuned here.

Optimal hyperparameters selected were: for TNG50 data set – truncation limit = 7, penalty discount = 15; for ALFALFA–NSA data set – truncation limit =

Table 4. Edge-count stability for the ALFALFA–NSA sample across the 3×3 hyperparameter grid centred on the adopted value ($t = 7$, $p = 35$). Each cell gives the total edge count, with the fully-directed-edge fraction in parentheses.

	$p = 28$	$p = 35$	$p = 42$
$t = 6$	14 (0.50)	13 (0.38)	12 (0.25)
$t = 7$	14 (0.50)	12 (0.25)	11 (0.27)
$t = 8$	12 (0.25)	11 (0.27)	10 (0.20)

Table 5. As Table 4, for the NSA sample ($t = 14$, $p = 50$).

	$p = 40$	$p = 50$	$p = 60$
$t = 13$	19 (0.32)	17 (0.29)	15 (0.27)
$t = 14$	17 (0.29)	16 (0.25)	14 (0.14)
$t = 15$	16 (0.25)	14 (0.14)	13 (0.15)

Table 6. As Table 4, for the TNG50 sample ($t = 7$, $p = 15$).

	$p = 12$	$p = 15$	$p = 18$
$t = 6$	40 (0.65)	37 (0.57)	34 (0.56)
$t = 7$	35 (0.49)	33 (0.48)	31 (0.45)
$t = 8$	33 (0.55)	33 (0.55)	30 (0.33)

7, penalty discount = 35; and for NSA data set – truncation limit = 14, penalty discount = 50.

To assess how strongly the recovered graphs depend on the chosen hyperparameters, FCIT was rerun on each real data set over a 3×3 grid centred on the adopted values, varying the truncation limit by ± 1 and the penalty discount by ± 20 per cent, with all other settings held fixed. Tables 4–6 report the total edge count, with the fraction of fully directed edges in parentheses, in each cell. These parenthetical values count only fully directed edges (both endpoints determined), whereas the oriented-edge counts reported in the comparative analysis below (for example 9 of 12 for the ALFALFA–NSA graph) also include edges carrying a single determined arrowhead, so the two measures differ. The total edge count varies by 33, 38, and 30 per cent across the grid for the ALFALFA–NSA, NSA, and TNG50 samples respectively, but the strongest physical relations—stellar mass downstream of luminosity in the observational samples, and the mass-to-kinematics edges in the simulation—remain present, and retain their baseline orientation, across most of the grid. The weaker, denser, and only partially oriented edges are more sensitive to the hyperparameter choice. The main comparative conclusions of the paper therefore remain robust to moderate changes in the settings, while the precise edge counts and densi-

ties should be regarded as accurate only at the ~ 30 per cent level.

A second source of variability is the FCIT search itself: the order-based search underlying FCIT (Best-Order Score Search / Greedy Relaxation of the Sparsest Permutation, BOSS/GRaSP) is not fully deterministic, so individual runs can differ in the orientation of some edges. To obtain a reproducible result I therefore ran FCIT 50 times on each of the ALFALFA–NSA and TNG50 samples, and 10 times on the larger NSA sample, and report the *consensus* graph: an edge is retained if it appears in at least half of the runs, and is assigned a single orientation only where one direction dominates in at least 60 per cent of the runs in which it appears (otherwise it is shown undirected). In practice the skeleton was completely stable—every reported edge appeared in all runs—and most orientations were stable as well, so the consensus graphs shown in Figs 4–6 are reproducible. In these figures an arrowhead marks an endpoint that FCIT determined to be an arrowhead, while an edge drawn without any arrowhead is undirected; I do not further distinguish a fully directed edge from a partially oriented one, since the tail mark is not stable across repeated runs. An arrowhead therefore encodes the relative ordering of the two variables rather than a fully resolved mechanism, and does not by itself exclude a latent common cause. All causal statements below refer to the ordering encoded in these consensus graphs.

3.3. Comparing causal structures across catalogues

Each matched sample of Section 2.6 is run through the same consensus procedure (50 FCIT runs; an edge kept at ≥ 50 per cent presence and oriented at ≥ 60 per cent dominance), at hyperparameters held fixed across all three simulations so that any difference between the recovered graphs reflects the data rather than the settings: truncation limit 7 and penalty discount 8. The consensus graphs are then compared edge by edge. For the three-code comparison each variable pair is classified by its *determined causal direction*—an arrowhead at a given endpoint counts as the same direction whether its tail is a circle or a bar, since the tail mark is not reproducible across runs. An edge is *invariant* if all three codes recover it with the same direction (the robust backbone), *conflicting* if it is present in all three but oriented differently, *majority* if present in two codes, and *code-specific* if present in only one.

4. RESULTS

4.1. ALFALFA–NSA Matched Sample

The PAG produced by applying the FCIT algorithm on the ALFALFA–NSA matched catalogue contains 12 edges (truncation limit = 7, penalty discount = 35). The PAG is shown in Fig. 4.

The result combines genuine relations among the gas-rich galaxies with imprints of the flux-limited, gas-selected sample.

Stellar mass (`ELPETRO_MASS`) lies at the end of several directed edges: it receives oriented links from baryonic mass, from r -band luminosity (`ELPETRO_ABSMAG_R`), and from the mass-to-light ratio (`ELPETRO_MTOL`). Because the stellar mass is itself derived from the photometry, FCIT places it downstream of the luminosity and mass-to-light information from which it is estimated, rather than treating it as a primary driver.

Both baryonic and stellar mass point to HI mass (`logMH`); these edges largely encode the definition of baryonic mass as the sum of stellar and HI mass. HI mass and half-light radius (`ELPETRO_TH50_R`) in turn point to redshift (`ZDIST`). In a flux- and gas-limited sample this orientation is the fingerprint of the selection: only the more HI-massive and larger galaxies are detectable at greater distances through Malmquist bias (Teerikorpi 2015), so distance becomes statistically predictable from these properties.

Colour and stellar metallicity (`ELPETRO_METS`) both point to the recent-to-past star-formation ratio (`ELPETRO_B300`), consistent with both tracking a galaxy’s star-formation history, while colour and the mass-to-light ratio remain linked without a resolved direction. The remaining undirected edges—between baryonic mass and r -band luminosity, and between HI line width (`W50`) and baryonic mass—are the residual mass–luminosity and Tully–Fisher (Tully & Fisher 1977) relations, whose direction FCIT cannot resolve from this narrow, gas-rich sample alone.

4.2. NSA Sample

The PAG produced by applying the FCIT algorithm on the NSA sample contains 16 edges (truncation limit = 14, penalty discount = 50). The PAG is shown in Fig. 5.

The graph is organised around r -band luminosity (`ELPETRO_ABSMAG_R`), which acts as a central hub: it receives oriented edges from colour, axis ratio (`ELPETRO_BA`), half-light radius (`ELPETRO_TH50_R`), and redshift (`ZDIST`), and in turn points to stellar mass (`ELPETRO_MASS`). The redshift→luminosity edge is the selection signature of the flux-limited sample, in which distance becomes predictive of luminosity; the remain-

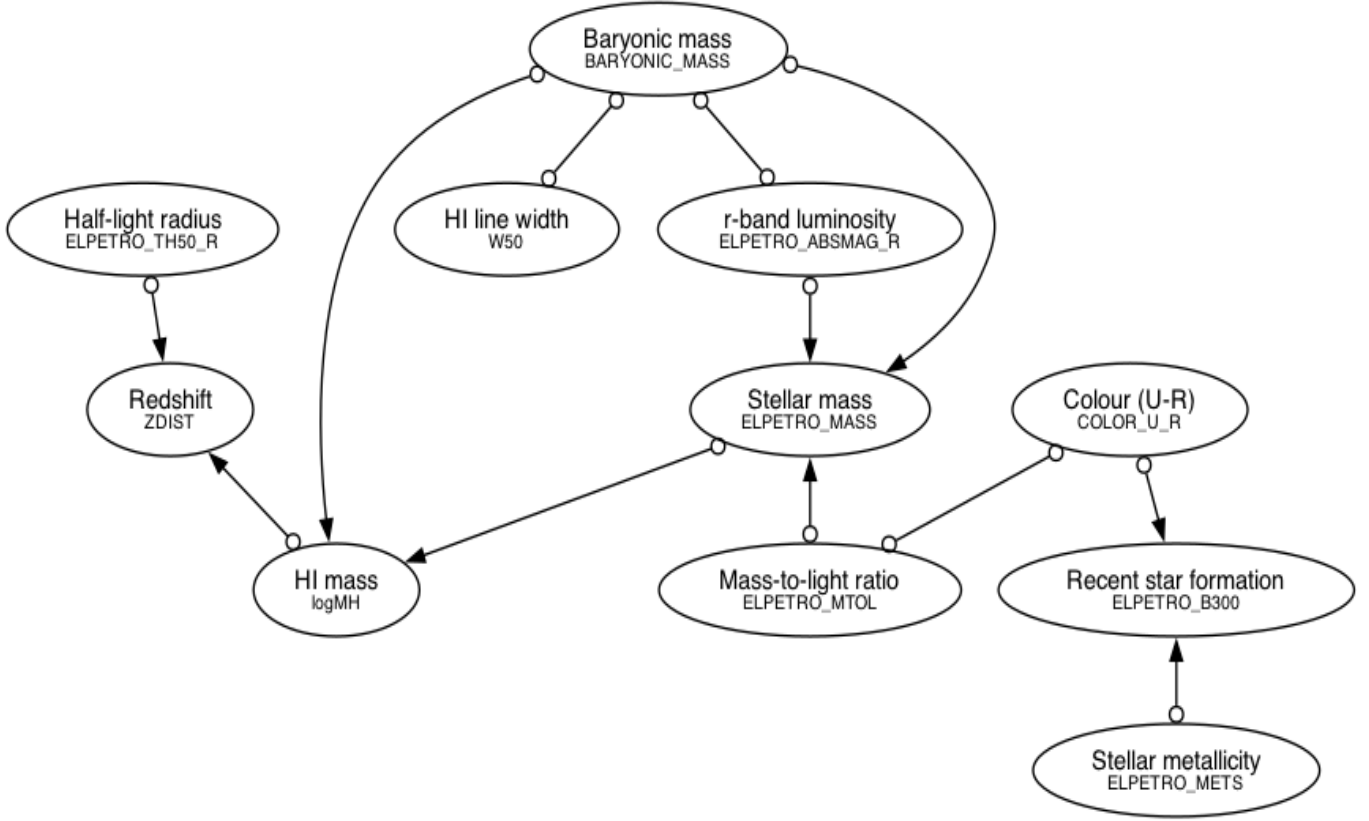


Figure 4. The PAG produced by the FCIT algorithm on the ALFALFA-NSA catalogue.

ing edges into luminosity recover the size–luminosity and colour–magnitude relations.

As in the gas-rich sample, stellar mass is a terminal sink: its only incoming edges are from luminosity and from the mass-to-light ratio (`ELPETRO_MTOL`), again reflecting that stellar mass is a photometrically derived quantity rather than an independent driver. This is an important caveat, since stellar mass is often described as the primary regulator of galaxy properties, whereas in the recovered graph it sits downstream of the photometric observables from which it is estimated.

The stellar-population variables—colour, the recent star-formation ratio (`ELPETRO_B300`), stellar metallicity (`ELPETRO_METS`), and the mass-to-light ratio—form a fully connected but undirected cluster, indicating strong mutual dependence through stellar age and metal content whose direction FCIT does not resolve. No direct stellar-mass–metallicity edge is recovered, so the mass–metallicity relation (Tremonti et al. 2004) appears here only indirectly, mediated by these population variables. The morphological quantities—Sérsic index, axis ratio, and half-light radius—form a second undirected group, with the Sérsic index also linked to colour and the half-light radius additionally linked to redshift; axis ratio and size feed the luminosity hub as noted above.

4.3. TNG50 Sample

The PAG produced by applying the FCIT algorithm to the TNG50 sample contains 33 edges (truncation limit = 7, penalty discount = 15). The PAG is shown in Fig. 6.

The TNG50 graph has a clear large-scale organisation. The kinematic variables—velocity dispersion (`VEL_DISP`) and maximum circular velocity (`VMAX`)—act as the principal sinks: 10 of the 17 oriented edges point into them, while the remaining oriented edges form the photometry and star-formation substructure described below. Stellar, baryonic, gas, and black-hole mass, half-mass radius, *u*-band luminosity, and stellar metallicity all have oriented edges into the kinematics, and velocity dispersion itself points to maximum circular velocity. Physically this is sensible: the mass and structure of a galaxy set the depth of its potential well, which determines its internal kinematics, so the kinematic quantities are recovered as downstream of mass.

By contrast, the mass variables themselves—dark-matter, stellar, baryonic, and gas mass—form a tightly connected but almost entirely *undirected* cluster. FCIT links them but cannot orient the edges between them, because they are strongly mutually correlated in the simulated population. Dark-matter halo mass (`DM_MASS`) is

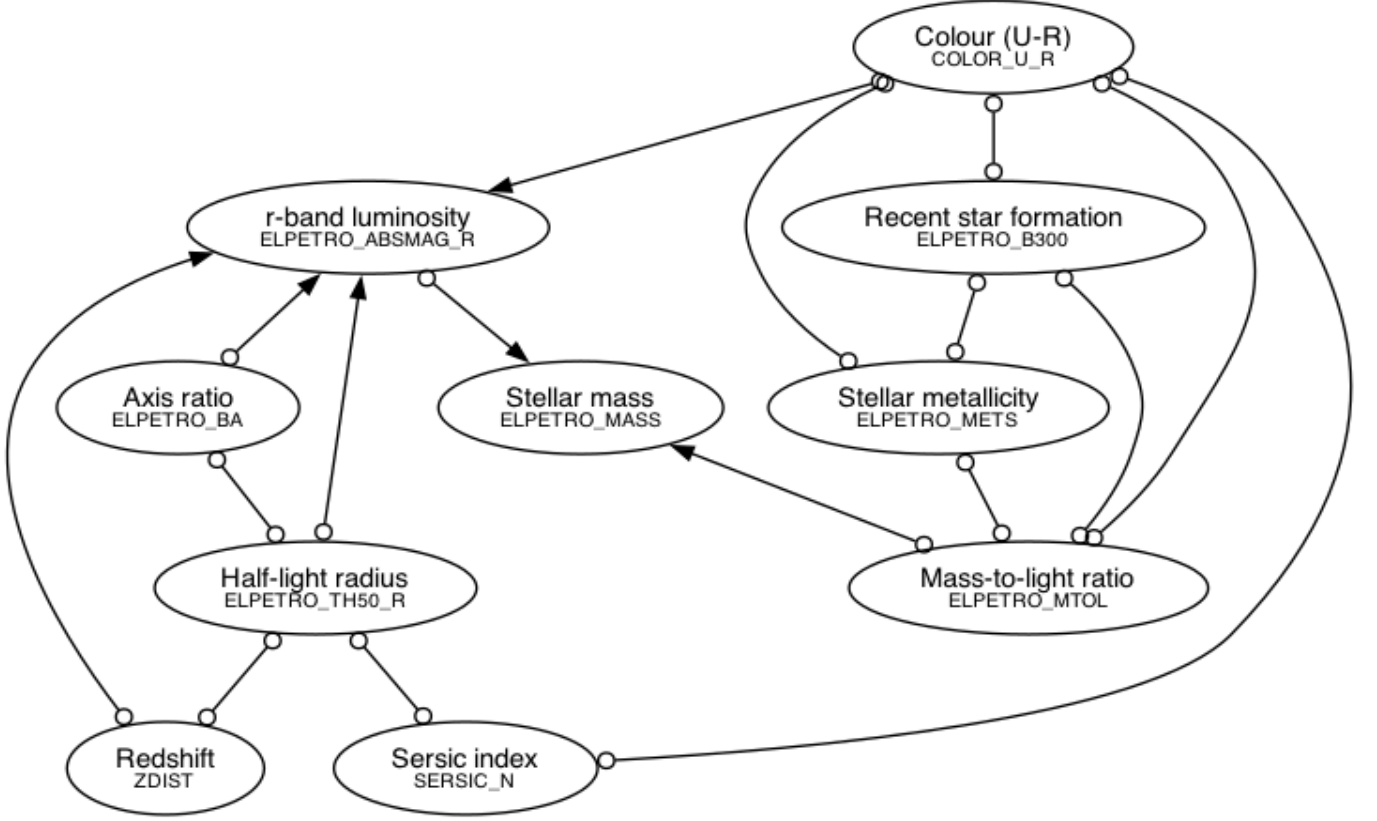


Figure 5. The PAG produced by the FCIT algorithm on the NSA catalogue.

the clearest example: its only two edges, to stellar mass and to colour, are both undirected, so no causal direction involving the halo is recovered at all. This is an honest and important negative result: despite halo mass being available, the near-collinearity of the mass variables prevents FCIT from placing the halo, or any single mass, at the head of a hierarchy.

A second, oriented sub-structure governs the photometry and star formation. Gas mass and colour both point to the r - and u -band luminosities, and gas mass, u -band, and r -band luminosity all point to the star-formation rate, consistent with the gas content and stellar populations driving a galaxy’s light and recent star formation. Black-hole mass points to both kinematic variables, reminiscent of the black-hole-velocity-dispersion relation; however, recall that two-thirds of the subhaloes have no resolved central black hole, so these edges should be treated as tentative. The gas-phase metallicity is only weakly connected, sharing a single undirected edge with stellar metallicity, and the two photometric bands are themselves linked without a resolved direction.

4.4. Comparative analysis

The three graphs recovered from the three samples show many differences. The number of edges rises from

the ALFALFA–NSA matched sample (12 edges) to the NSA sample (16 edges) and finally to the TNG50 cosmological simulation sample (33 edges).

This reflects the physical differences between the samples: the ALFALFA–NSA sample contains a specific set of galaxies—blue, gas-rich objects—resulting in the fewest edges. The NSA sample covers a broader set of galaxies but lacks some properties as an observational survey, leading to a moderate number of edges. The TNG50 sample includes the full range of galaxies and all their properties, yielding the most edges.

Since algorithms like FCIT depend on data variation to detect conditional dependencies, greater variation in galaxy properties leads to denser graphs. Normalising by the number of possible variable pairs, the corresponding edge densities are 0.15, 0.36, and 0.36 for the ALFALFA–NSA, NSA, and TNG50 samples respectively; the density is therefore lowest for the gas-selected sample and comparable for the optical and simulated samples, so the larger raw edge count of TNG50 partly reflects its larger set of variables rather than a proportionally denser graph.

Dark-matter halo mass is the primary property missing from the observational samples, where it acts as an unmeasured confounder and likely contributes to their

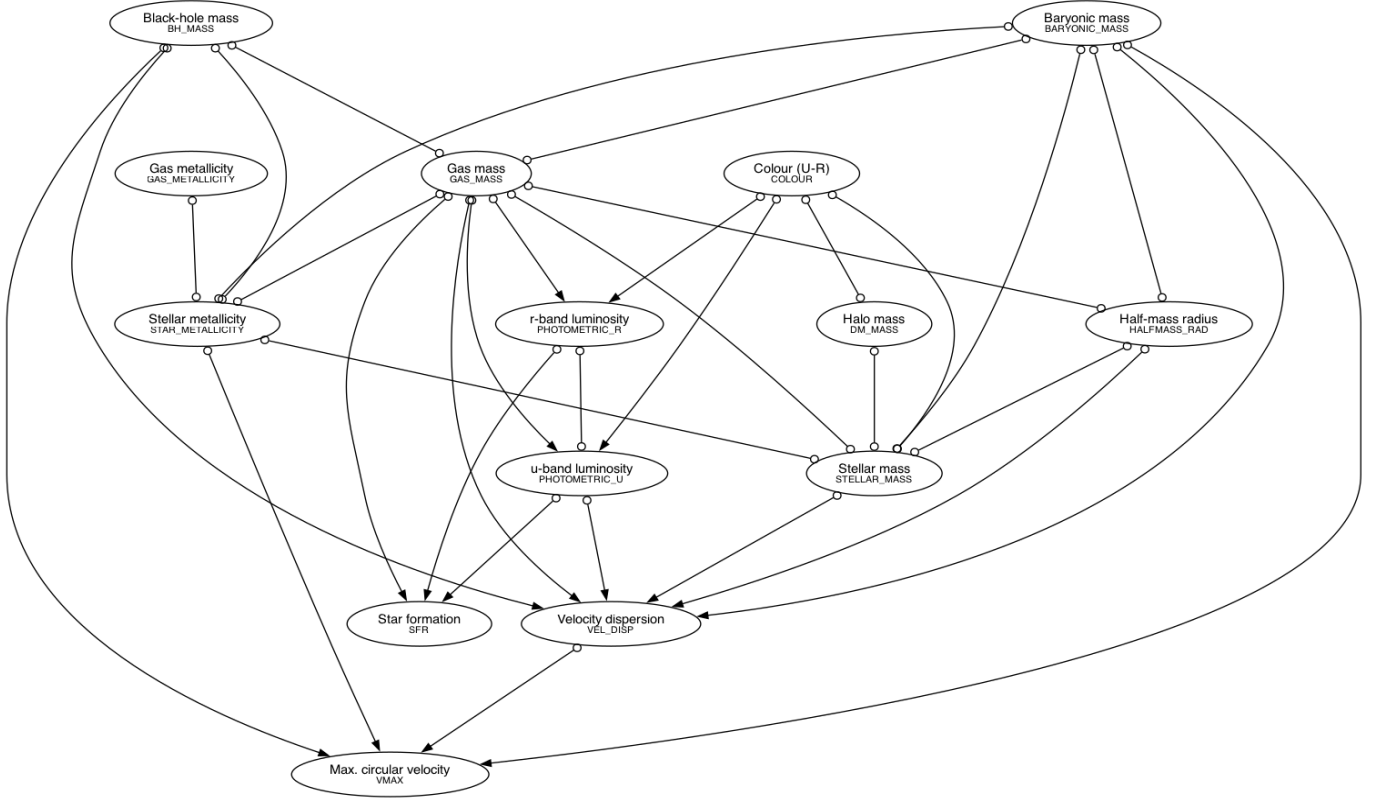


Figure 6. The PAG produced by the FCIT algorithm on the TNG50 cosmological sample.

partially oriented and undirected edges. Its inclusion in TNG50 does not, however, translate into a halo-centred hierarchy: as described above, the halo and the other mass variables form an undirected cluster, and the oriented edges instead run from mass and structure into the kinematic variables.

A common pattern emerges across all three samples. FCIT confidently orients edges where there is a genuine asymmetry—the selection edges that make distance predictable from observed properties, the derived-quantity edges by which stellar mass follows from the photometry, and the mass-to-kinematics edges in the simulation—but leaves tightly correlated groups of variables (the stellar-population indicators in the observational samples, the mass variables in the simulation) undirected. The fraction of edges carrying a determined arrowhead is accordingly highest in the compact, selection-dominated ALFALFA–NSA graph (9 of 12 edges) and lower in the more heterogeneous NSA and TNG50 graphs (6 of 16 and 17 of 33 respectively).

The ALFALFA–NSA sample has nearly twice as many galaxies as the TNG50 sample, yet its PAG is the sparsest, which suggests that the diversity of physical variation in a sample, rather than its size, is an important driver of the richness of the recovered causal structure.

I caution, however, that the three samples were also run at different penalty discounts and contain different variable sets, both of which affect edge density, so this attribution is indicative rather than fully controlled.

5. CAUSAL STRUCTURE ACROSS INDEPENDENT SIMULATIONS

The per-sample graphs above are each recovered on a catalogue’s own variable set and at its own settings, so they cannot be compared directly. I now compare causal structure on the matched samples of Section 2.6, following the procedure of Section 3.3.

5.1. The invariant backbone and the model-dependent relations

The decisive comparison uses the full intrinsic set all three simulations provide, where halo mass, kinematics, and gas content are free of the derived-quantity problem of the observations. The three codes share the scaling relations they are calibrated to, but use independent solvers and feedback models. Their consensus graphs contain 39, 29, and 31 edges for TNG50, EAGLE, and SIMBA (Table 7). Of the 56 union edges, just eight are recovered with the same direction by all three—a penalty-dependent count of four to eight (Table 9)—while eight more are present in all three but oriented

Table 7. Agreement of the recovered causal structure across the three independent simulations, over the union of 56 edges in the matched intrinsic graphs.

Class	Edges
Invariant (same direction in all three)	8
Conflicting (all three, different direction)	8
Recovered by two codes	11
Recovered by one code	29
Union	56

differently, 11 are found by two codes, and 29 by one. Fig. 7 shows the 16 edges common to all three: black where the direction agrees, orange where it differs.

The eight invariant edges form a physically sensible backbone (Fig. 7, black), keyed by stellar mass into velocity dispersion—a mass–velocity–dispersion, Faber–Jackson-like relation (Faber & Jackson 1976). That all three codes recover it supports causal discovery recovering genuine physics where the physics is comparatively clear, though the codes share calibration targets, so their agreement is not fully independent confirmation. Most other invariant edges are fixed by construction: the mass–bookkeeping links from baryonic and stellar mass to gas mass, and the algebraic photometric links among colour, the bands, and star formation. The robust core is therefore essentially what gravity and the variable definitions fix in common, while the divergent edges (orange) are those set by the subgrid model. I caution, however, that the three simulations differ in mass range, resolution, and measurement aperture (the EAGLE quantities use a 30-kpc aperture), so part of this divergence—the EAGLE mass–luminosity reversal in particular—may reflect these measurement differences rather than feedback physics alone. The clearest conflict is the mass–luminosity relation, oriented stellar mass to luminosity by TNG50 and SIMBA but the reverse by EAGLE; the mass–metallicity relation is oriented only by EAGLE, and the stellar–halo relation by TNG50 and EAGLE but left undirected by SIMBA. The black hole–velocity dispersion relation (Gebhardt et al. 2000) is recovered by two codes in opposite directions, but rests on a floored variable (two-thirds of subhaloes lack a resolved black hole) and is tentative.

5.2. Is the model-dependence real, or pipeline noise?

A low cross-code agreement is only meaningful if it exceeds the variability of the procedure itself, and cross-code *agreement* is reassuring only if the orientations are also *accurate*. I therefore ran two tests. First, a within-code null test separating two questions—whether the codes recover the same *set* of relations (the skeleton) and whether they orient shared relations the same way.

I drew two *disjoint* $N = 5,000$ subsamples of each simulation (sharing no galaxies) and, at this matched sample size, measured both the within-code (two draws of one code) and cross-code (one draw of each) agreement at equal statistical power, repeating the split over 40 independent random partitions to place an error budget on the contrast. Skeleton agreement is the Jaccard overlap of the edge sets; orientation agreement is the fraction of edges *both* graphs orient that receive the same direction. Second, an orientation-*accuracy* test: I applied the identical consensus procedure to the 100 latent-free mock datasets with known causal graphs used to tune the hyperparameters (Section 3) and, of the true edges FCIT orients, measured the fraction given the correct direction.

The orientations FCIT commits to are *accurate* (Table 8): across the 100 mocks, of the true edges it orients, 85 per cent receive the correct direction (per-mock mean 0.86, 95 per cent bootstrap CI 0.82–0.89; chance 0.50). The *skeletons*, by contrast, are strongly model-dependent, and this contrast is robust to the particular split: over 40 independent random partitions the within-code skeleton Jaccard has median 0.79 (95 per cent interval 0.58–0.93) against a cross-code median of 0.33 (0.24–0.52), and a permutation test rejects their equality at $p < 0.001$. The within-code *orientation* agreement is likewise high (median 0.92), but the cross-code orientation agreement (median 0.75) rests on only 6–11 edges, and its 95 per cent interval (0.40–1.00) includes the 0.50 chance level, so it is *not* established as above chance; a gold-standard SIMBA check (two independent 10,000-galaxy draws) gives 0.93. The two firm results are therefore the 85 per cent orientation *accuracy* against known graphs and the strongly model-dependent skeleton.

The picture is therefore sharper than a blanket “model dependence.” Causal discovery orients what it recovers both accurately (against known graphs) and consistently (across codes); where the codes diverge is on *which* relations appear. Part of that skeleton variation is real model dependence (cross-code median 0.33, well below the within-code 0.79; permutation $p < 0.001$), but part is pipeline noise, since even a single code reproduces only about 0.79 of its own skeleton on independent data. The invariant backbone is correspondingly soft—4–8 edges depending on the penalty (Table 9)—and mostly definitional.

Two conclusions follow. First, the causal structure recovered from a single simulation is only partly transportable: the codes recover substantially different *sets* of relations (skeleton overlap median 0.33 against a within-code 0.79; permutation $p < 0.001$), a caution for single-dataset causal-discovery studies (Desmond & Ramsey

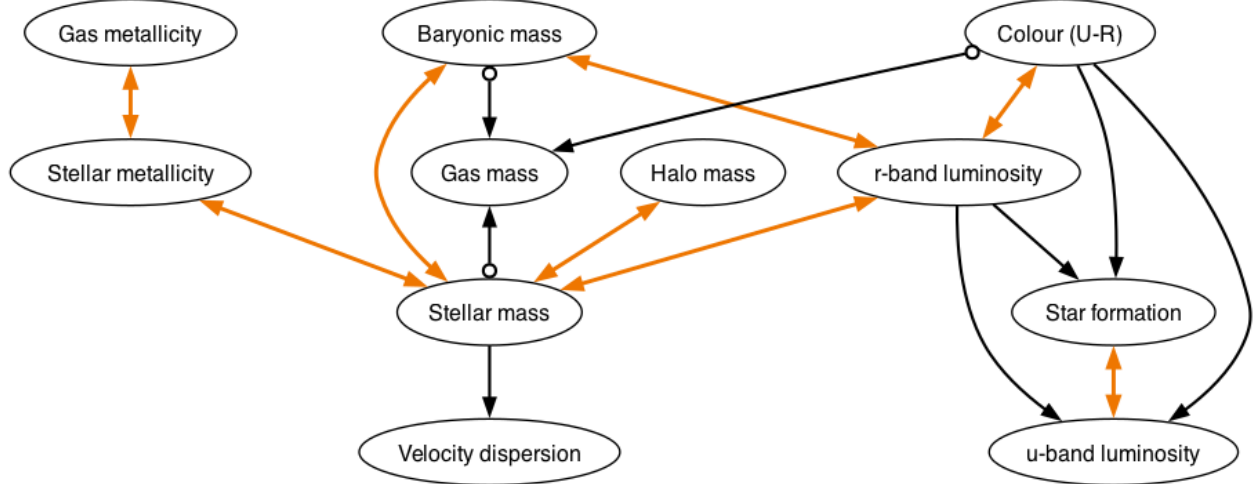


Figure 7. The sixteen causal edges recovered by *all three* independent simulations on the matched intrinsic variables. Black edges have the same direction in all three codes—the invariant backbone of eight edges (a largely definitional core plus the mass–velocity–dispersion relation), keyed by stellar mass $\rightarrow$ velocity dispersion. Orange double-headed edges are present in all three codes but oriented differently between them (model-dependent direction). Edges recovered by only one or two codes are omitted and counted in Table 7.

Table 8. Within-code (two disjoint draws) and cross-code agreement of the recovered causal structure, separated into skeleton (Jaccard overlap of the edge sets) and orientation (fraction of edges oriented by *both* graphs that receive the same direction; the number of such edges is in parentheses). Chance-level orientation agreement is 0.50. All within- and cross-code rows use disjoint $N = 5,000$ draws (matched sample size), except the fourth, a gold-standard pair of disjoint $N = 10,000$ SIMBA draws. Orientation *accuracy* against the known mock graphs ($t=7$, $p=8$) is 0.85. A permutation test over 40 random partitions confirms the within-versus-cross skeleton contrast (median 0.79 vs 0.33, $p < 0.001$); the rows below show one representative partition, while medians and intervals over the 40 partitions are quoted in the text.

Comparison	Skeleton (Jaccard)	Orientation agreement
Within TNG50	0.74	1.00 (11)
Within EAGLE	0.61	0.83 (12)
Within SIMBA	0.82	0.88 (17)
Within SIMBA (10,000)	0.77	0.93 (14)
TNG50 vs EAGLE	0.32	0.67 (6)
TNG50 vs SIMBA	0.31	0.83 (6)
EAGLE vs SIMBA	0.47	0.64 (11)

Table 9. Dependence of the cross-code comparison on the penalty discount p . Per-code edge counts and the union grow as p falls, but the number of edges present in all three codes with the *same* direction (the invariant backbone) stays small at every penalty.

p	Edges (TNG50/EAGLE/SIMBA)	Union	Present in all 3	Invariant
6	48 / 37 / 38	68	15	4
8	39 / 29 / 31	56	16	8
12	32 / 24 / 25	50	10	4

2026; Jin et al. 2025; Pasquato et al. 2023; Bluck et al. 2022). Second, and more positively, the *orientations* FCIT assigns are accurate (85 per cent correct against known mock graphs, versus 50 per cent chance), so causal structure separates a robust, orientable core from the model-dependent remainder that the shared scaling relations cannot probe.

6. DISCUSSION AND LIMITATIONS

6.1. Interpretation of the Recovered Causal Structures as Physical Diagnostics

The three recovered partial ancestral graphs highlight the fact that causal structure is an astrophysical diagnostic which varies with sampling criteria and the completeness of variable measurements.

In the sparsest graph, from the ALFALFA–NSA data set, much of the oriented structure is set by the selection function and by the way quantities are derived, rather than by gas physics. HI mass and half-light radius point to redshift—the signature of a flux- and gas-limited sample—and stellar mass is recovered downstream of the photometric quantities (baryonic mass, luminosity, mass-to-light ratio) from which it is estimated. The genuinely physical relations that survive, such as the Tully–Fisher (Tully & Fisher 1977) and mass–luminosity links, are present but left undirected. A picture of gas-regulated star formation (Lilly et al. 2013; Peng & Maiolino 2014) is thus consistent with, but not strongly demonstrated by, this graph, because the narrow, gas-rich dynamic range leaves little variation for FCIT to exploit.

The NSA graph, of intermediate complexity, is organised around r -band luminosity rather than stellar mass. Luminosity is the hub into which colour, size, axis ratio, and redshift feed, and from which stellar mass in turn follows; the stellar-population indicators—colour, the star-formation ratio, metallicity, and mass-to-light ratio—form a strongly connected but undirected cluster (Gallazzi et al. 2005; Brinchmann et al. 2004). The often-quoted role of stellar mass as the primary regulator of galaxy properties is therefore *not* recovered: because stellar mass is itself a photometric estimate, FCIT consistently places it downstream of the observed light. This is a useful reminder that a causal-discovery result reflects how each quantity is measured as much as the underlying physics.

In the TNG50 graph, the explicit inclusion of halo mass and kinematics reveals a different limitation. The oriented edges form a coherent picture in which mass and structure determine the internal kinematics—velocity dispersion and maximum circular velocity act as sinks fed by stellar, baryonic, gas, and black-hole

mass and by half-mass radius—while gas content and colour drive the photometry and star formation. The mass variables themselves, however, including the dark-matter halo, are so tightly correlated that FCIT leaves the edges among them undirected; the expected mass–metallicity relation (Ma et al. 2016; Torrey et al. 2019) likewise appears only as undirected links among the mass and metallicity variables, and the halo acquires no oriented edge at all. The simulation therefore does not recover a halo-down hierarchy. Rather than a failure, this is informative: it shows that, with realistically collinear mass variables, even a complete data set does not allow constraint- and score-based discovery to resolve the direction of the mass–mass relations—a caution that applies equally to the observational samples, where stellar populations play the same role. Bursty star formation (Tacchella et al. 2020) and dust (Calzetti et al. 2000) likely contribute further to the photometric and star-formation edges remaining only partially resolved.

In conclusion, these results suggest that causal structure is an important astrophysical diagnostic, varying according to the sample selection criteria and completeness of variables.

6.2. Limitations

There are several important limitations worth mentioning. Firstly, the use of all causal discovery algorithms is conditioned by the Markov assumption, faithfulness and acyclicity (Spirtes et al. 2000; Pearl 2009). Acyclicity in particular is a strong assumption for feedback-regulated galaxies, in which star formation, gas, and AGN activity plausibly form closed loops; genuinely cyclic mechanisms would surface as unstable or conflicting orientations across codes, which may itself contribute to some of the divergences reported here.

Secondly, observational samples lack direct measurements of important latent variables like dark matter halo mass and dust attenuation which produce numerous circle end points. In contrast, although TNG50 addresses this problem, it introduces modelling errors.

Thirdly, mock data calibration cannot perfectly replicate true observational noise and biases; moreover the calibration mocks are latent-free acyclic graphs, whereas the motivation for PAGs is latent confounding, and the penalties were tuned to a skeleton-recovery (F1) score rather than an orientation-recovery score, while the NSA hyperparameters were imported from Desmond & Ramsey (2026) without independent calibration here. Fourthly, the analyses are performed using only a limited number of first-order galaxy properties. Fifthly, the hyperparameter sensitivity analysis on real data, although now reported across a 3×3 grid, still spans only

a limited neighbourhood of the chosen values. Sixthly, the black-hole mass is dominated by subhaloes with no resolved central black hole (two-thirds sit at the resolution floor), so it violates the continuous-input assumption of the basis-function independence test and can perturb any conditioning set it enters, not only the edges incident on it; the edges involving it are therefore only tentative, and a cleaner treatment would restrict that variable to galaxies with a resolved black hole, or drop it altogether. Dropping it entirely, however, leaves the invariant backbone unchanged (eight invariant edges, sixteen present in all three codes; per-code edge counts 36/28/28), so the headline cross-code result does not depend on it. Seventhly, the cross-code comparison uses a common standardised variable set with dust-free photometry, so it measures agreement on the *direction* of each relation rather than its precise strength. Eighthly, the observation–simulation comparison is limited to the few properties measured comparably in both; a forward-modelled treatment is the natural next step. Ninthly, the three simulations are read from three different public databases with non-identical mass apertures and variable definitions (the EAGLE quantities, for example, use a 30-kpc aperture); standardisation equalises statistical power but not these definitional differences, so part of the cross-code divergence—EAGLE’s in particular—could reflect measurement heterogeneity rather than physics alone, and a matched-aperture reanalysis is the natural check.

7. CONCLUSIONS

Using the FCIT algorithm I recover the partial ancestral graphs of galaxy properties in two observational catalogues (NSA and ALFALFA–NSA) and three independent cosmological simulations (TNG50, EAGLE, and SIMBA), and compare them on matched variables, developing causal discovery into a test of galaxy-formation models. The main conclusions are:

1. Causal discovery orients the relations it recovers reliably: the stellar mass–velocity dispersion (Faber–Jackson) direction (Faber & Jackson 1976) is recovered identically by all three codes, the key-stone of a small invariant backbone (four to eight edges, depending on the penalty), most of which is fixed by the variable definitions.
2. The model-dependence is concentrated in the skeleton rather than the orientation. Over 40 random partitions the recovered edge sets overlap only weakly across codes (skeleton Jaccard median 0.33, against 0.79 within a code; permutation $p < 0.001$)—a robust, significant gap. The orientations of shared edges agree more often within a

code (median 0.92) than across (median 0.75), but the cross-code figure rests on few edges and is not established above the 0.50 chance level. Which relations a simulation recovers is therefore partly a fingerprint of its subgrid physics.

3. The orientations FCIT assigns are accurate, not merely self-consistent: on 100 latent-free mock datasets with known causal graphs, 85 per cent of the edges it orients receive the correct direction (chance 50 per cent). Because a single code nonetheless reproduces only about 0.79 of its own skeleton on independent data, part of the cross-code skeleton difference is intrinsic pipeline variability rather than physics.
4. This cautions single-dataset causal-discovery studies—the *set* of relations recovered need not transport to another code—while the reliability of the orientations makes causal structure a diagnostic that discriminates galaxy-formation models on multivariate structure their shared scaling relations cannot probe.

The approach extends naturally to larger simulation suites and, with forward-modelled observations, to a like-for-like comparison between real and simulated galaxies. In short, causal discovery tests not only whether a simulation reproduces the properties of galaxies, but whether it reproduces the way those properties determine one another.

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DATA AVAILABILITY

The NASA–Sloan Atlas (NSA) v1.0.1 catalogue is publicly available at <https://www.sdss4.org/dr17/manga/manga-target-selection/nsa/>. ALFALFA survey data and documentation are available at <https://egg.astro.cornell.edu/alfalfa/data/>, and the IllustrisTNG (TNG50) simulation data are accessible at <https://www.tng-project.org/data/docs/specifications/>. The EAGLE simulation data are available through the public database (McAlpine et al. 2016) at <http://virgodb.dur.ac.uk:8080/Eagle/>, and the SIMBA CAESAR catalogues

at <http://simba.roe.ac.uk/>. The ALFALFA–NSA cross-match used in this work is not publicly released but can be recreated following the procedure in Stiskalek

et al. (2021). The code used in this paper is available at <https://github.com/vidigoat/astrophysics>.

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